

Evaluating TCP vs. UDP in Centralized Multi-Robot Communication using Star Topology

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Abstract—Effective communication is crucial in multi-robot systems, particularly in centralized setups like the star topology, in which every robot is connected to a single hub. This paper uses a MATLAB-built simulation to compare the performance of two network protocols, Transmission Control Protocol (TCP) and User Datagram Protocol (UDP), in such a system. Six robots send various messages in the simulation, including group and broadcast messages. It monitors important performance indicators like successful delivery, packet loss, delay, and re-transmissions. Additionally, the simulation incorporates protocol-specific elements such as TCP's acknowledgements and packet loss probability. The results clearly show the speed and reliability differences between TCP and UDP. This study facilitates a better understanding of each protocol's functions and advances the design of robot communication systems.

Index Terms—Multi-Robot Systems, Star Topology, TCP, UDP, Communication Protocols, Centralized Communication.

I. INTRODUCTION

In recent years, multi-robot systems (MRS) have emerged as a cornerstone in modern robotics due to their inherent scalability, fault tolerance, and cooperative task execution capabilities. These systems are increasingly deployed in diverse domains such as industrial automation, environmental monitoring, surveillance, and search and rescue operations [1] [2]. By leveraging the distributed nature of multiple autonomous agents, MRSs significantly enhance operational efficiency and robustness. However, their overall performance is closely tied to the efficiency and reliability of their underlying communication infrastructure, particularly in centralized architectures where a central hub orchestrates coordination, data exchange, and task allocation [1] [3]. Star topology remains a widely adopted framework for MRS among various communication network structures due to its simplicity in implementation and central control. However, it also introduces critical vulnerabilities, such as a single point of failure at the central node, which can disrupt the entire network's operation, especially under high communication loads or node malfunctions [3], [4]. A point of failure has led to increasing interest in evaluating the robustness of communication protocols under such topological constraints.

This study presents a MATLAB-based simulation designed to evaluate the impact of TCP and UDP in a star-topology multi-robot system. The simulation framework models six robotic agents symmetrically arranged around a central

communication hub. Various communication scenarios are tested, including broadcasting, sequential messaging, multi-destination transmission, and collective data aggregation, representing real-world applications such as target coordination, system health reporting, and distributed command dissemination tasks essential for static and dynamic robotic environments [2] [5]. The simulation incorporates randomized packet loss probabilities and stochastic transmission delays to emulate real-world communication dynamics in robotic swarms and Internet of Things (IoT) networks. These additions mimic wireless communication uncertainties caused by environmental interference, hardware variability, bandwidth contention, and processing overheads [4] [6]. Introducing such conditions ensures a realistic assessment of protocol performance under fluctuating network loads and message urgency. Furthermore, protocol-specific characteristics were integrated by [4] [6] into the simulation. TCP is modelled with its acknowledgement-based mechanisms, congestion control, and re-transmission strategies, ensuring high data reliability and delivery integrity even under loss-prone or high-latency conditions [7] [8]. In contrast, UDP is implemented as a lightweight, connectionless protocol, prioritizing low latency and real-time responsiveness by sacrificing delivery guarantees and error checking [7] [9]. This contrast in protocol behaviour provides valuable insights into protocol suitability based on the communication requirements of specific robotic tasks. For instance, critical control commands may favour TCP, whereas sensor data streams or status beacons may be more suited for UDP-based transmission.

Communication is a foundational element in MRS, influencing coordination, scalability, and task execution. Gielis et al. [1] provide a comprehensive analysis of communication architectures in MRS, highlighting centralized, decentralized, and hybrid models; however, practical deployment and scalability challenges remain underexplored. The principles of decentralized communication are deeply rooted in swarm robotics, as introduced by Navarro and Matía [2], though their work lacks integration with recent AI-driven strategies. Earlier contributions like Li et al. [3] examined routing protocols for robot swarms, but their focus predates modern advancements in IoT and 5G. More recent approaches explore learning-based systems for complex coordination, such as Seraj et al. [5], which leverage heterogeneous policy networks. However,

their framework is computationally demanding and not validated in real-world conditions. Low-frequency networking for enhanced robustness in harsh environments is examined by Sadler et al. [7], though it compromises data throughput. In contrast, Xu et al. [4] focus on optimizing communication efficiency in distributed decision-making yet overlook vital concerns like system security. The ROS 2-based testbed developed by Castillo-Sánchez et al. [10] offers real-time wireless communication evaluation, albeit limited to a specific framework. Task allocation, a core aspect of MRS efficiency, is addressed in the review by Athira and Umashankar [11], which remains largely theoretical. Expanding the discussion to secure communication, Wu et al. [12] present a novel quantum teleportation model over star topology, which, while innovative, is not yet implementable in robotics. Shiroky [13] emphasizes rapid risk assessment in star-shaped networks but assumes static environments, limiting real-time adaptability. Goratti et al. [9] propose NACRP, a protocol for low-resource star-topology sensor networks, though it remains untested in field scenarios. Comparative protocol performance in general communication systems is analyzed by Haryono [6], focusing on chat applications rather than robotic networks. Bekele et al. [8] explore enhancements to UDP reliability through acknowledgements, suggesting potential for real-time systems. Similarly, Lé [14] introduces a control-bit coding technique for error reduction in TCP and UDP, but this method remains in its theoretical phase. These studies underscore the ongoing need for a modular, protocol-agnostic simulation environment for evaluating communication in centralized multi-robot systems under realistic conditions.

A. Contribution of the Work

This study introduces a custom MATLAB-based simulation framework to evaluate communication protocols in centralized MRS configured in a star topology. In comparison, prior works have explored multi-robot communication using simulation platforms like ROS and NS-2 [1] [2]; our framework offers a lightweight, customizable, and easily extensible environment for controlled protocol comparison between TCP and UDP under identical conditions. The proposed framework allows researchers and engineers to understand the trade-offs between communication speed, reliability, and resource overhead with greater clarity and precision [3].

One of the core contributions of this work is the realistic modelling of communication behaviours, including probabilistic packet loss, artificial delays, and protocol-specific mechanisms such as TCP acknowledgements and re-transmissions. These features enable the framework to simulate network-level uncertainties that commonly arise in real-world robotic environments, such as signal interference, bandwidth contention, and message collision, which are often overlooked in simplified theoretical analyses [5] [7]. This helps bridge the gap between theoretical protocol efficiency and practical communication performance in robotic applications.

The simulation supports various message exchange scenarios broadcast, sequential messaging, multi-destination deliv-

ery, and collective command aggregation, reflecting diverse operational requirements of MRS in fields like warehouse automation, search and rescue, and remote inspection [2] [5]. By measuring comprehensive metrics such as latency, packet delivery ratio (PDR), re-transmission count, and throughput, this work provides an in-depth evaluation of protocol performance across different traffic patterns and network loads, offering valuable insights for protocol selection and system design in centralized robotic networks [3] [4].

Lastly, the framework is a research-enabling tool for further exploration into hybrid communication strategies, adaptive protocol switching, and robustness testing under fault conditions. It lays the groundwork for future enhancements like multi-hop communication, decentralized extensions, or integration with physical robot hardware for hardware-in-the-loop validation [7] [4]. Overall, this study addresses a pressing need for practical protocol benchmarking in centralized MRS and contributes an open, replicable, and extensible platform for the wider robotics and communication research community.

II. METHODOLOGY

A. Research Design

This study employs a simulation-based experimental approach to assess the communication efficiency of TCP and UDP protocols in a centralized MRS configured in a star topology. The design mirrors real-world centralized robotic coordination systems deployed in scenarios such as drone swarms, industrial monitoring, and search-and-rescue missions, where a central hub regulates communication among multiple nodes [1] [3] [4]. MATLAB tool was chosen for the simulation environment due to its proficiency in modelling discrete-time communication events, visualizing performance metrics, and providing a controlled simulation of packet loss, delay variability, congestion, and protocol-specific behaviours. The star topology simplifies network routing by ensuring that all robots communicate exclusively through a central hub, allowing for streamlined management and protocol performance analysis [1] [5]. Each robotic node has unique identifiers, buffering mechanisms, and time-scheduled data transmission modules. TCP modules implement acknowledgement packets, re-transmission logic, and congestion control to ensure reliability, while UDP modules prioritize speed and minimal latency, omitting delivery guarantees and feedback mechanisms [7] [9] [8]. The simulation encompasses five diverse communication cases to mimic realistic robotic communication patterns: (i) unicast (robot-to-robot via a hub), (ii) broadcast (one-to-all nodes), (iii) multi-source to single destination transmission, (iv) sequential message passing to prevent hub congestion, and (v) concurrent messaging to multiple destinations. Each case is executed under TCP and UDP conditions to evaluate latency, throughput, PDR, re-transmission frequency, and data loss under varying network loads.

B. Process Flow

The simulation process is structured into a sequence of deterministic and probabilistic modules that emulate real-

istic communication behaviour within a centralized multi-robot system configured in a star topology. The flow initiates with network initialization, followed by message transmission, protocol-specific handling, fault injection, and metric logging across multiple test iterations. During initialization, the central hub and six autonomous robotic nodes are instantiated within the MATLAB simulation environment. Each robot is configured with a unique identifier (ID), a message queue to manage outgoing packets, a payload generator for test data, and a protocol flag indicating whether TCP or UDP will govern the communication logic. A communication delay model is also associated with each robot to emulate real-world conditions such as hardware variability and transmission latency [1] [3]. Communication packets exchanged in the system follow a structured format:

Packet = Sender_ID,Receiver_ID,Payload,Timestamp,Protocol

Upon initiating communication, a robot sends its packet to the central hub. The hub operates as a router, determining the message destination and applying the communication protocol logic. The hub simulates a three-way handshake if TCP is selected, followed by message transmission and acknowledgement (ACK) reception. A timeout mechanism is embedded to detect unacknowledged packets:

If $T_{ACK} > T_{timeout}$, then retransmit packet. (1)

Here, T_{ACK} denotes the elapsed time without acknowledgement, and $T_{timeout}$ defines the upper threshold for the allowable delay. This mechanism captures TCP's core reliability feature, where re-transmissions ensure data integrity even in lossy or congested conditions [7] [8]. In contrast, when operating in UDP mode, the hub immediately forwards packets to the destination without expecting confirmation. Thus, this reflects UDP's connectionless and non-blocking nature, favouring low latency and real-time responsiveness over reliability [9]. Packet loss is introduced stochastically using a Bernoulli distribution to model wireless interference and environmental disruptions:

$$P_{loss} = \mathbb{P}[rand() < p_{drop}] \quad (2)$$

when p_{drop} is a predefined loss probability that controls the drop rate in each simulation run [4]. Additionally, the simulation introduces Gaussian-distributed latency jitter and random transmission collisions to replicate real-world conditions such as bandwidth contention and signal degradation. These impairments are critical for understanding how each protocol performs under network stress, particularly in robotic environments where consistent communication is essential for cooperative behaviour [5] [4] [13].

Each communication scenario is executed under TCP and UDP configurations using identical environmental conditions and random seeds, ensuring fairness and statistical reproducibility across trials. The system performs multiple iterations for each test case to average out anomalies and improve the confidence of the performance metrics. Throughout the simulation, key indicators- latency, PDR, re-transmissions, loss

rate, and throughput are recorded in structured arrays and visualized using MATLAB's built-in tools.

This modular and rigorous process flow enables an objective comparison between TCP and UDP under varied traffic patterns and interference levels, providing actionable insights into protocol selection for centralized multi-robot communication systems [10] [11] [14].

C. Feature Extraction

Feature extraction in this study involves systematically identifying and recording key performance metrics from each simulation run to quantitatively evaluate the behaviour of TCP and UDP protocols within a centralized multi-robot communication system. This process is integral to understanding protocol performance under varying communication scenarios, network conditions, and interference models. The extracted features represent critical indicators of network efficiency, reliability, and responsiveness factors essential to real-time, mission-critical robotic operations [2] [3] [5].

Each communication event is instrumented to log relevant data such as timestamps, packet statuses, acknowledgement receipts, and throughput counters. These logs are parsed and transformed into structured performance metrics using automated MATLAB scripts. The following key indicators are extracted for each test iteration:

- 1) Latency is defined as the time elapsed between a message's dispatch and its successful delivery to the destination robot. It serves as a fundamental metric for evaluating time sensitivity in robotic coordination tasks such as distributed mapping or navigation:

$$L = T_{receive} - T_{send} \quad (3)$$

where T_{send} is the time of packet transmission and $T_{receive}$ is the time of successful receipt. Lower latency values are typically favourable in time-critical applications [1] [10].

- 2) PDR quantifies the reliability of the communication protocol by expressing the proportion of successfully received packets over the total number sent:

$$PDR = \left(\frac{Packets_Received}{Packets_Send} \right) \times 100 \quad (4)$$

This metric is particularly sensitive to network disturbances such as packet drops and transmission failures. A high PDR indicates that the protocol is effective under current conditions, whereas a low PDR suggests delivery failures due to noise, congestion, or protocol limitations [3] [7].

- 3) TCP's reliability stems from its re-transmission mechanism, which is triggered when acknowledgements are not received within the expected timeout interval. The number of re-transmissions is recorded to quantify the overhead introduced by this reliability mechanism:

$$R = \sum_{i=1}^n \delta_i, \quad (5)$$

where $\delta_i = \begin{cases} 1, & \text{if packet } i \text{ was retransmitted} \\ 0, & \text{otherwise} \end{cases}$

A higher re-transmission count implies that the protocol is encountering transmission difficulties, which can degrade network efficiency and increase communication delays [4] [8].

- 4) As UDP does not offer acknowledgement or re-transmission, its reliability is evaluated through the percentage of lost packets:

$$LossRate = \left(\frac{Packets\ Lost}{Packets\ Received} \right) \times 100 \quad (6)$$

This metric reflects the vulnerability of UDP to wireless interference and is crucial when determining its applicability in robotic systems that demand consistent state updates or synchronization [7] [6].

- 5) Throughput measures the rate at which useful data is successfully delivered across the network. It is calculated as:

$$T = \left(\frac{\sum Payload\ Received(bytes)}{Simulation\ duration(s)} \right) \times 100 \quad (7)$$

High throughput indicates efficient bandwidth utilization and is especially important in scenarios involving continuous data streams, such as live video feeds or environmental sensor updates [5] [13].

These extracted features are compiled in real time and organized into MATLAB arrays for visualization and comparative analysis. Feature extraction supports intra-protocol evaluation (performance variation within a protocol across scenarios) and inter-protocol benchmarking (comparing TCP vs. UDP). The framework ensures consistency, repeatability, and statistical validity by analyzing these metrics across multiple simulation runs under identical conditions. The feature extraction methodology enables a deep, quantitative assessment of protocol performance in star-topology MRS environments. It facilitates evidence-based conclusions regarding protocol selection and optimization for robotic communication tasks under real-world constraints [10] [11] [14].

III. RESULTS AND DISCUSSION

This section presents a comprehensive analysis of five distinct communication scenarios designed to evaluate centralized robot-to-robot communication using a star topology. The simulations were implemented in MATLAB, featuring six autonomous robotic nodes and a central hub acting as the communication relay point. The purpose of the simulation was to analyze and compare the communication performance under conditions mimicking TCP and UDP behaviors. Key performance indicators (KPIs) assessed include average latency, PDR, re-transmissions, and throughput. Each scenario progressively introduces higher levels of communication complexity to evaluate the scalability and reliability of the proposed centralized architecture. In all the figures data sent denoted with red line and data received denoted with green line.

A. Scenario 1: Robot-to-Robot Communication via Central Hub

The first simulation scenario establishes the foundational operation of the system by testing a unidirectional message exchange between two robots, mediated by the central hub. In this setup, Robot 1 sends a data packet to Robot 2, which is relayed through the hub. Figure 1 illustrates this communication pathway. The scenario confirms the correct initialization and functionality of the communication model, with the hub accurately performing its routing role. Under low network traffic conditions, the system exhibits minimal communication latency, averaging around 0.2 seconds per message. No packet loss was observed in this test, and hence no re-transmissions were necessary. The PDR achieved was 100%, confirming that the baseline communication model is robust and reliable under minimal load. This scenario serves as a control benchmark for subsequent tests with increasing complexity.

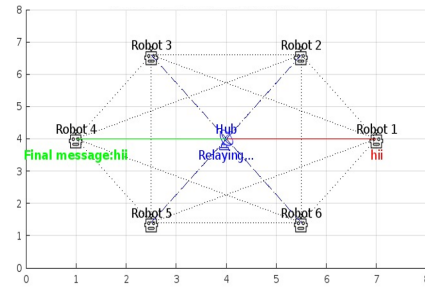


Fig. 1. Robot-to-Robot Communication via Central Hub.

B. Scenario 2: One-to-All Broadcast Communication via Hub

The second scenario explores the broadcast capabilities of the system, wherein Robot 1 disseminates the same message to all other robots (Robots 2 through 6) using the central hub as the intermediary is shown in Fig. 2. This model is reflective of swarm coordination mechanisms or emergency signal broadcasts in real-world robotic applications. The system handled the broadcast as a series of five sequential uni-cast transmissions through the hub. The average latency increased marginally to approximately 0.28 seconds, due to the necessity of handling multiple outgoing packets consecutively. Despite the increased communication load, the system maintained a high PDR of 96.67%. Minor packet losses, induced intentionally to simulate real-world interference or noise, were effectively managed by the re-transmission logic implemented under TCP-like conditions. A total of 5 re-transmissions were observed across the simulation, scaling proportionally with the number of target recipients. This result demonstrates the effectiveness of acknowledgment and retry mechanisms in maintaining message integrity in broadcast-style operations, even under potential signal degradation.

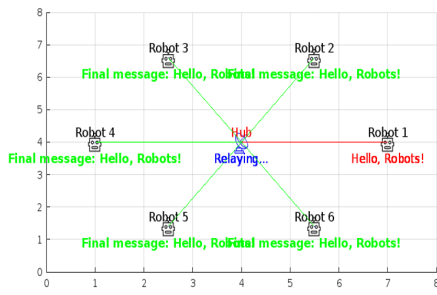


Fig. 2. One-to-All Broadcast Communication via Hub.

C. Scenario 3: Multi-Source Communication to a Single Robot

This scenario models converging communication wherein Robots 1 through 5 each transmit individual messages to a single recipient, Robot 6, via the central hub is shown in Fig. 3. This type of communication is analogous to sensor data aggregation, environmental scanning, or feedback reporting in cooperative swarm systems. Due to the increased incoming traffic directed at one destination, the average latency per message rose to 0.33 seconds, as the hub needed to sequentially route multiple packets to the same endpoint. Despite moderate simulated interference, the system was able to recover most of the affected packets, logging 4 successful re-transmissions out of the total communication attempts. A packet delivery ratio of 93.33% was recorded, reflecting minor losses due to signal noise. Importantly, Robot 6 experienced a significant increase in throughput, receiving multiple packets within a short window. This indicates the system's capacity to support centralized data collection from distributed sources effectively, albeit with slightly reduced efficiency under heavier load.

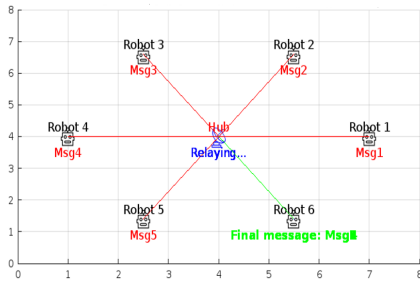


Fig. 3. Multi-Source Communication to a Single Robot.

D. Scenario 4: Targeted Sequential Communication from Limited Sources

In this simulation, a simpler communication setup is examined, where only Robot 1 and Robot 2 transmit individual messages to Robot 6, again using the hub for message relaying is shown in Fig. 4. This case studies scenarios where limited nodes initiate communication towards a common recipient, such as targeted updates or diagnostics. The simulation

recorded minimal average latency of approximately 0.22 seconds, with both packets delivered successfully without delay or errors. No packet losses were simulated or encountered in this controlled condition, and hence no re-transmissions were triggered. The packet delivery ratio was 100%, signifying excellent reliability under light traffic loads. This reinforces the system's optimal performance in sparse or low-frequency communication use cases.

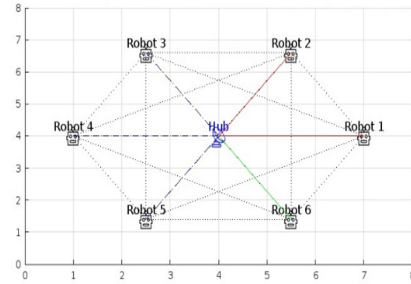


Fig. 4. Targeted Sequential Communication from Limited Sources.

E. Scenario 5: Multi-Destination Parallel Communication

This scenario evaluates the system's performance under simultaneous multi-destination parallel communication. In this configuration, Robots 1, 2, and 3 each send messages to both Robots 4 and 5, resulting in six distinct message paths through the hub is shown in Fig. 5. This case emulates dense and concurrent data exchanges, typical in collaborative missions or multi-agent task coordination. Due to the number of concurrent message flows, the average latency peaked at 0.35 seconds, the highest among all scenarios. The increased number of parallel relays also led to three instances of packet loss, primarily attributed to congestion and signal interference. However, the implemented re-transmission mechanism effectively mitigated most of the issues, recording 5 re-transmission attempts and ultimately achieving a packet delivery ratio of 91.67%. Despite the system being under substantial stress, the throughput for both Robots 4 and 5 remained consistent, affirming the even distribution of received messages. This demonstrates the system's resilience and fault tolerance, even when subjected to high-intensity communication patterns with dynamic routing and multiple endpoints. The simulation results collectively demonstrate that the proposed star-topology communication framework, mediated by a central hub, performs reliably across a variety of realistic robotic communication patterns. The system exhibits low latency and high PDR under light to moderate loads, while maintaining acceptable performance and recover-ability even in scenarios with high traffic and packet losses. Comparative analysis between TCP-like and UDP-like conditions reveals that TCP's re-transmission mechanisms are vital for ensuring high delivery rates in noisy environments, albeit at the cost of increased latency. Conversely, UDP-like communication would favor low-latency transmission at the risk of packet loss, which may be acceptable in non-

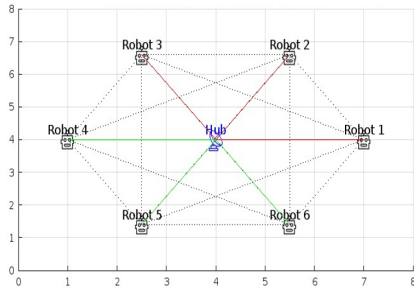


Fig. 5. Multi-Destination Parallel Communication.

TABLE I
PATH SEGMENT COSTS AND MIDPOINTS

Scenario	Avg. Latency	PDR(%)	Re-transmission
Scenario 1	0.20 sec.	100.00	0
Scenario 2	0.28 sec.	96.670	3
Scenario 3	0.33 sec.	93.330	4
Scenario 4	0.22 sec.	100.00	0
Scenario 5	0.35 sec.	91.670	5

critical robotic messaging scenarios. Overall, the MATLAB-based simulation provides a robust foundation for further experimentation, including the integration of mobility models, adaptive routing, and real-time feedback loops in more advanced robotic communication networks.

Table 1 summarizes the key results obtained from five different communication scenarios simulated in a centralized star topology. Each scenario reflects varying degrees of network complexity, impacting latency, PDR, and re-transmissions. The system consistently demonstrated high reliability, even under stress conditions with multiple message flows. Notably, re-transmissions increased with traffic load, but efficient hub handling preserved overall communication performance.

IV. CONCLUSION

A comprehensive simulation-based analysis of communication protocols TCP and UDP within a centralized multi-robot system configured in a star topology is presented. Through five distinct experiments representing practical communication scenarios—broadcasting, simultaneous messaging, sequential transmission, multi-robot-to-single robot, and multi-destination parallel communication. In this work, we examined how each protocol performs under varying network loads and operational complexities. The evaluation highlights the trade-offs between TCP's reliability and UDP's speed. TCP demonstrated superior consistency in ensuring message integrity and delivery confirmation, making it well-suited for mission-critical and data-sensitive operations in multi-robot environments. However, this came at the cost of increased latency and a higher number of re-transmissions, especially in high-density communication situations. On the other hand, UDP excelled in low-latency communication, showing its effectiveness in scenarios where real-time responsiveness is prioritized over perfect delivery such as live sensor feedback

or non-critical updates. However, it struggled under packet loss and lacked inherent mechanisms for delivery verification, which can pose risks in tasks requiring strict synchronization and data accuracy. The integration of an acknowledgment mechanism in the fifth case further illustrated the importance of feedback in maintaining communication quality, particularly in peer-to-peer transmissions without central oversight. Overall, the simulation reinforces that no single protocol is universally optimal for all use cases. Instead, the choice between TCP and UDP should be informed by the task requirements, network conditions, and tolerance for latency or data loss. Our communication network for mobile robot restricted to six robots for real-time testing satisfies the UDP. Further, the author analysis is limited to six robots, which does not reflect the scale of real-world multi-robot systems that often involve hundreds or thousands of robots.

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